JAMA | Review

Survivors of Cyanotic Congenital Heart Disease
A Review

Gabriele Egidy Assenza, MD; Edward O'Leary, MD; Margarita Brida, MD; Michael Kwon, MD; Michael Job Landzberg, MD

IMPORTANCE Cyanotic congenital heart diseases, which occur in approximately 0.2% of live births in the United States, have high mortality rates if untreated, but survival to adulthood is common with current surgical interventions.

OBSERVATIONS Approximately 80% of all cyanotic congenital heart diseases are caused by tetralogy of Fallot (TOF), D-loop transposition of the great arteries (TGA), and congenital heart disease with single-ventricle circulation. Survivors of cyanotic congenital heart diseases benefit from multidisciplinary care including pediatric and adult cardiologists, congenital cardiac surgeons, and electrophysiologists. Ninety percent of patients with TOF survive more than 30 years after surgery to close the ventricular septal defect and repair the right ventricular outflow tract. Nearly all adults with repaired TOF develop right ventricular volume overload due to pulmonary regurgitation, and atrial tachycardias and atrial fibrillation occur in approximately 20% to 45% of patients by age 45 years. For D-loop TGA, which involves the aorta arising from the right ventricle and the pulmonary artery arising from the left ventricle, arterial switch procedures result in survival rates of 93% to 97% to age 30 years. After atrial switch operation, 30% to 50% of patients develop moderate or severe right ventricle dysfunction by age 25 years; atrial tachycardia occurs in 48% to 63% of patients at 32 to 40 years postsurgery, and sinus node dysfunction is common. Sudden cardiac death occurs at a mean age of 30 to 35 years (SD, 6.4 years) in up to 15% of adults who have undergone an atrial switch operation. Infants born with a single ventricle that supplies both systemic and pulmonary circulation are most often treated with staged open-heart surgical interventions, typically performed during a period from neonatal age to 6 years of age, culminating in the Fontan procedure, which connects the inferior and superior vena cava to the pulmonary arteries, allowing deoxygenated blood to flow to the lungs without a pumping ventricle. Survival rates for children who undergo the Fontan procedure are 50% to 80% at age 40 to 50 years, although these patients may develop New York Heart Association functional class III or IV (0.35% per person-year) and have increased risk of early death or heart transplant requirement (0.36% per person-year).

CONCLUSIONS AND RELEVANCE With surgical intervention, survival to adulthood is common among patients with TOF, D-loop TGA, and single ventricle. However, these survivors of cyanotic congenital heart diseases are at risk of valve dysfunction, arrhythmias, heart failure, and premature death. Optimal care involves multidisciplinary management including pediatric and adult cardiologists, congenital cardiac surgeons, and electrophysiologists.

JAMA. 2026;336(1):56-68. doi:10.1001/jama.2026.8806
Published online June 8, 2026.

Multimedia

CME at jamacmelookup.com

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: Michael Job Landzberg, MD, Palliative Therapeutics in Congenital Heart Disease (PATCH), Boston Adult Congenital Heart (BACH) and Pulmonary Hypertension Program, Department of Cardiology, Boston Children's Hospital, 300 Longwood Ave, Boston, MA 02115 (mike.landzberg@cardiochboston.org).

Congenital heart diseases (CHDs) are the most common type of major structural birth defects, affecting 0.86% to 1.0% of all live births; cyanotic CHDs occur in approximately 0.2% of live births in the United States.[1] The diagnosis of CHD is usually made prenatally based on routine anatomy ultrasonography performed between 18 and 22 weeks of pregnancy. However, occasionally CHDs are diagnosed postnatally. Tetralogy of Fallot (TOF; ≈0.35% of all live births), D-loop transposition of the great arteries (TGA; ≈0.3%), and CHD with single-ventricle circulation (≈0.4%) are the most common cyanotic CHDs, causing approximately 80% of all cyanotic CHDs (**Box**).[1]

Cardiac-related cyanosis results from reduced pulmonary blood flow and right to left shunt with TOF (**Figure 1**A), parallel circulation (Figure 1B) with D-loop TGA, or intracardiac mixing of systemic and pulmonary venous return with single ventricle (Figure 1C).[2] After surgical correction, 90% of patients with TOF survive more than 30 years.[3] Among infants with D-loop TGA, 90% survive beyond age 40 years with modern arterial switch operation, although atrial switch procedures confer only 60% to 82% survival to age 40 years.[4,5] Overall survival for patients with single ventricle after staged Fontan procedure is 50% to 80% at age 40 to 50 years.[6-11] Adults with cyanotic CHD are at increased risk of heart failure, arrhythmias, and valve dysfunction.[12] Multidisciplinary care is required to reduce long-term morbidity and mortality.[13,14]

This Review discusses the pathophysiology of cardiovascular complications in adults with TOF, D-loop TGA, and single-ventricle, lesion-specific congenital and surgical anatomy after early management and also discusses late complications and management strategies (Box).

Box. Questions Regarding Cyanotic Congenital Heart Disease

What are the most common cyanotic congenital heart diseases?
The 3 most common cyanotic congenital heart diseases are tetralogy of Fallot, transposition of the great arteries, and single-ventricle congenital anomalies.

What are the most common long-term cardiovascular complications in survivors of cyanotic congenital heart disease?
Heart failure and arrhythmias are the most common long-term adverse cardiovascular consequences of cyanotic congenital heart disease. Clinical management of heart failure in this patient population may require surgical or interventional cardiology approaches to repair anatomic abnormalities, such as valvular regurgitation or stenosis. Treatment of arrhythmias, most commonly intra-atrial reentrant tachycardia and slow-conducting isthmuses-dependent ventricular tachycardia, is typically accomplished by cardiologists with expertise in clinical electrophysiology and adult congenital heart disease.

What should clinicians know about lifetime management of survivors of cyanotic congenital heart disease?
Optimal care of adult survivors with repaired cyanotic congenital heart disease requires close collaboration between generalist clinicians and comprehensive care centers with pediatric and adult congenital cardiologists, congenital cardiac surgeons, electrophysiologists, transplant teams, hepatologists, and mental health clinicians. Multidisciplinary care is recommended to reduce morbidity and mortality.

Methods

We performed a PubMed search for English-language articles, using the search terms *transposition of the great arteries, Fontan, single ventricle, ACHD heart failure, systemic right ventricle heart failure, Fontan circulatory failure, ACHD heart transplant*, and *tetralogy of Fallot*, published between January 1, 2015, and December 31, 2025. We prioritized high-quality randomized clinical trials and observational studies with large sample sizes and included practice guidelines, narrative reviews, and consensus statements, if appropriate. Of 10 949 retrieved articles, we included 113 studies consisting of 5 consensus documents, 5 meta-analyses, 9 narrative reviews, 74 observational studies, 4 practice guidelines, 6 randomized clinical trials, 2 systematic reviews, and 8 case reports.

Pathophysiology

Cyanosis is defined as a bluish discoloration of the skin and mucous membranes caused by an increased concentration of deoxygenated hemoglobin (typically exceeding 5 g/dL) in the capillary circulation. Persistent hypoxemia is associated with hematologic abnormalities (secondary erythrocytosis, hyperviscosity, and iron dysregulation), endothelial dysfunction, and increased risk of thromboembolic and hemorrhagic complications and can cause kidney impairment, hepatic fibrosis or cirrhosis, reduced exercise capacity, and neurocognitive effects such as anxiety or attention disorders.[15] Cardiac-related cyanosis results from reduced pulmonary blood flow and right to left shunt as in TOF (Figure 1A), parallel circulation (Figure 1B) with D-loop TGA, or intracardiac mixing of systemic and pulmonary venous return with single ventricle (Figure 1C).[2]

Among survivors of surgically repaired CHD, the complex interplay between congenital anatomic factors; surgical interventions, such as incisions and suturing into cardiac muscle; intracardiac or extracardiac rerouting of blood flow; valvular regurgitation; residual shunt; increased ventricular afterload; and pulmonary vascular disease results in compensatory activation of the sympathetic nervous system, the renin-angiotensin-aldosterone system, and arginine vasopressin. Over time, these effects increase the risk of heart failure and atrial and ventricular arrhythmias due to ventricular hypertrophy, dilation, and fibrosis and atrial pathology (**Table 1** and **Figure 2**).[14,16-25] End-stage heart failure may require heart transplant, but many of these patients have extracardiac organ disease (primarily kidney, hepatic, and pulmonary) that may lead to consideration of multiorgan transplant.[25,26]

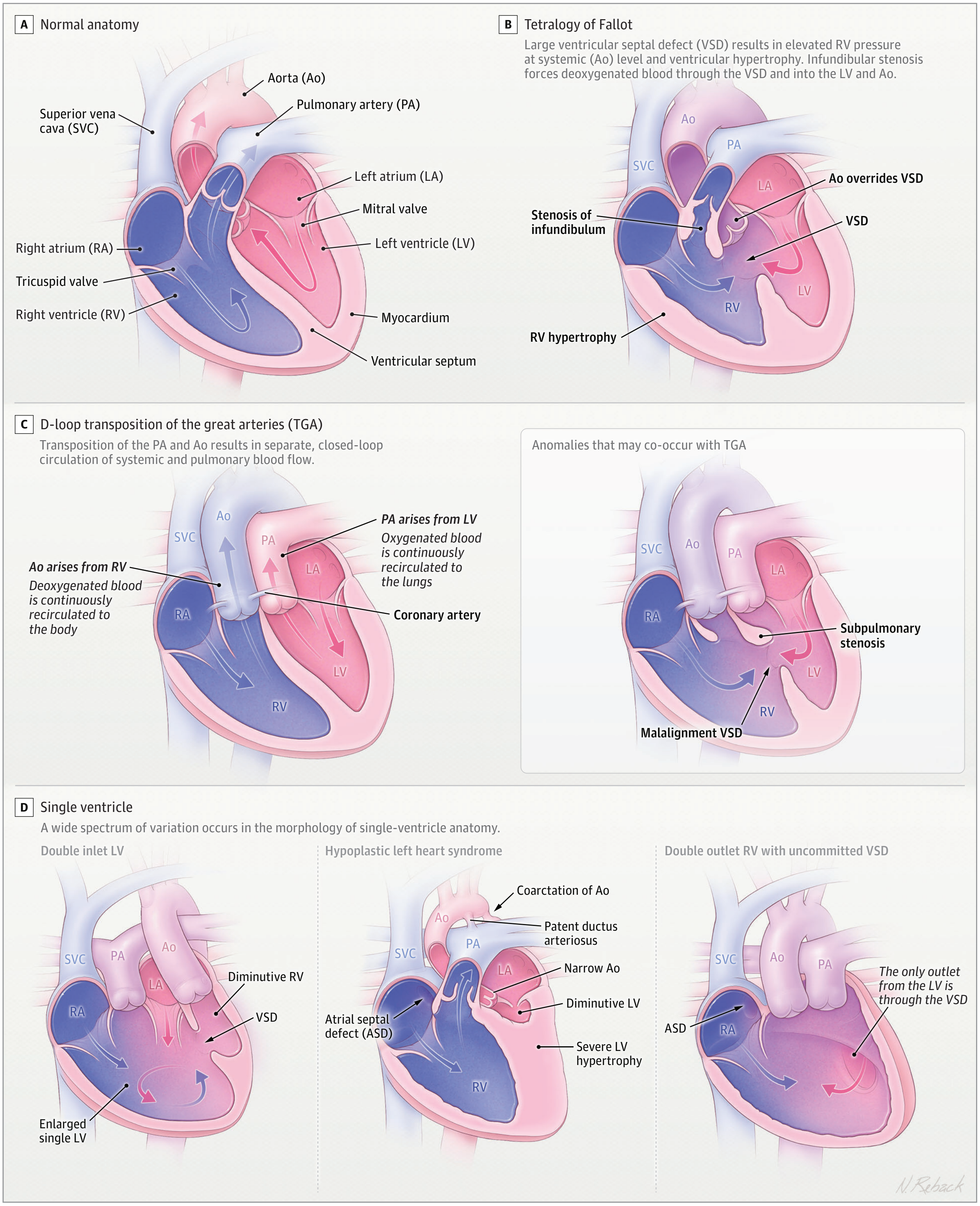
Clinical Presentations of Cyanotic CHDs

TOF

Anatomy and Early Management

The 4 main features of TOF are right ventricular (RV) outflow tract obstruction, a large ventricular septal defect, an aorta that overrides the ventricular septal defect so that it receives blood from both the right and left ventricular (LV) chambers, and RV hypertrophy (Figure 1A).[27] Cyanosis severity depends on the degree of RV outflow tract obstruction affecting pulmonary blood flow and the amount of right to left shunting across the ventricular septal defect. Primary corrective surgery consists of closure of the ventricular septal defect and widening of the RV outflow tract with a patch,

Figure 1. Illustration of Anatomy of Common Cyanotic Congenital Heart Diseases



Principal anatomic features of tetralogy of Fallot, VSD, overriding Ao, RV hypertrophy, and infundibular stenosis due to conal septum deviation. Wide anatomic spectrum of single-ventricle heart includes true univentricular heart, severe ventricular hypoplasia, or complex congenital heart defect not amenable for biventricular repair, such as double outlet RV with uncommitted or remote VSD, straddling atrioventricular valves, and atrial isomerism (D).

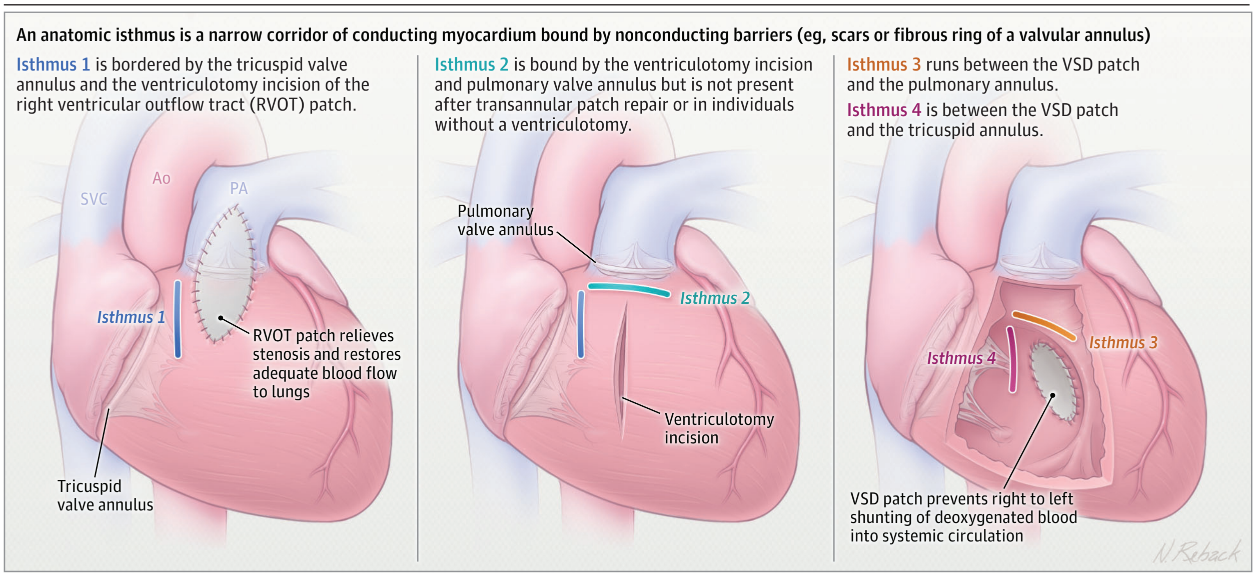
Table 1. Risk Factors for Ventricular Tachycardia and Sudden Cardiac Death in Select Congenital Heart Disease Anatomies

Risk factors for VT/SCD	Tetralogy of Fallot[14,21,22]	TGA after atrial switch[16]	Fontan[17-20]
Electrophysiologic	QRS duration >180 ms	Wide QRS duration	History of pacemaker
	QRS fractionation	Atrial arrhythmias	Nonsustained VT
	Anatomic isthmus conduction velocity <0.5 m/s	Nonsustained VT	
	Inducible VT with programmed ventricular stimulation	Lack of β-blockade	
	Atrial arrhythmias		
Hemodynamic	LVEF <55%	RVEF <35%	
	RVEF <45%	LV outflow tract gradient >36 mm Hg	
	RV late gadolinium enhancement extent on CMR	History of congestive heart failure	
	RV mass to volume ratio >0.45 g/mL	≥Moderate systemic tricuspid regurgitation	
	LV EDP ≥12 mm Hg		
	Peak oxygen uptake ≤17 mL/kg/min		
	B-type natriuretic peptide ≥127 pg/mL		
Surgical	Prior palliative shunt		Stage 1 with RV-PA conduit
	Ventriculotomy incision		Atriopulmonary Fontan
	Surgical repair before 1980		
Miscellaneous	Older age	Older age	Older age
	Syncope	Syncope	

Abbreviations: CMR, cardiac magnetic resonance; EDP, end-diastolic pressure; LVEF, left ventricular ejection fraction; PA, pulmonary artery; RVEF, right ventricular EF; SCD, sudden cardiac death; TGA, D-loop transposition of the great arteries; VT, ventricular tachycardia.

SI conversion factor: To convert B-type natriuretic peptide to ng/L, multiply by 1.0.

Figure 2. Illustration of the 4 Anatomic Isthmuses That May Serve as Substrate for Monomorphic Ventricular Tachycardia in Tetralogy of Fallot[a]

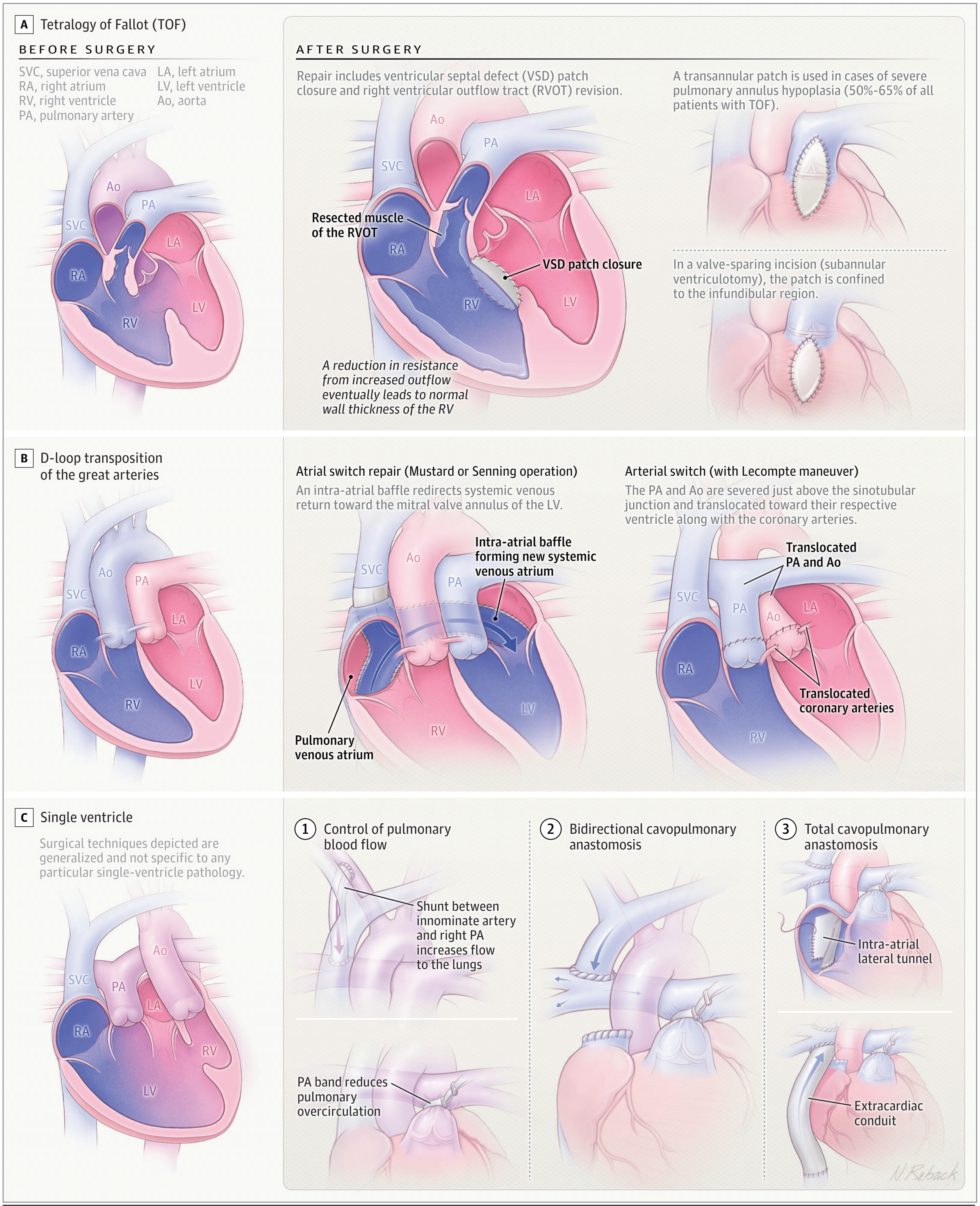


Prototypical example of arrhythmia mechanism in a repaired cyanotic congenital heart disease. The propensity to propagate ventricular tachycardia depends on slow conduction velocity, which results from progressive fibrosis. Ao indicates aorta; PA, pulmonary artery; SVC, superior vena cava; and VSD, ventricular septal defect.

[a]Modified from Kapel et al.[23]

which is recommended before age 3 to 6 months (**Figure 3**A). The RV outflow tract patch may require incision of the pulmonary valve annulus (transannular extension), resulting in pulmonary regurgitation that causes progressive RV volume overload and dilation, affecting 40% to 85% of patients within 5 to 10 years after repair and nearly all patients with transannular patch RV outflow tract repair during their lifetimes.[28,29] Pulmonary regurgitation leads to progressive RV dilation, resulting in reduced RV ejection fraction and myocardial fibrosis and scarring. These RV changes can impair LV filling, thereby causing LV systolic and diastolic dysfunction.[29] In a meta-analysis of 7 retrospective cohort studies (2854 patients with mean age 28 years [SD, 4]), LV dysfunction occurred in up to 20% to 25%

Figure 3. Illustration of Surgical Repair of Common Cyanotic Congenital Heart Disease



A, Standard surgical repair of tetralogy of Fallot includes VSD closure and RVOT revision using either transannular patch or valve-sparing approach. B, Surgical repair of D-loop transposition of the great arteries: atrial switch repair (Mustard or Senning) redirects systemic and pulmonary venous return to the PA and Ao, respectively, restoring physiologic streaming without resolving ventriculoarterial discordance, which the arterial switch operation restores. C, Surgical steps of single ventricular palliation: first stage controls pulmonary blood flow, avoiding either pulmonary overcirculation or severe cyanosis; second stage: surgical bidirectional cavopulmonary anastomosis between the SVC and the ipsilateral PA ensures controlled, low-pressure pulmonary blood flow, reducing ventricular volume; final stage of total cavopulmonary anastomosis reroutes the inferior VC to the PA, using either an extracardiac conduit or an intra-atrial (lateral tunnel) baffling, leading to near-normalization of systemic saturation and ventricular loading.

Table 2. Supporting Evidence and Potential Concerns Regarding Commonly Used Heart Failure Drugs for Survivors of Cyanotic Congenital Heart Disease

Drug	Supporting evidence in ACHD: TOF	TGA arterial switch	sRV (TGA atrial switch)	SV (Fontan)	Potential concerns
Diuretics					Renal insufficiency Gout flare Electrolyte imbalance Symptomatic hypovolemia
Loop	EO	EO	EO	EO	
Thiazides	EO	EO	EO	EO	
β-Blocker	NA	NA	Potential harm[35]	NA	Bradycardia Negative inotropic effect Negative effect on conduction system
ACE inhibitor	RCT[36,a]	NA	NA	NA	Hypotension Electrolyte imbalance Renal insufficiency
ARBs	NA	NA	EO[37]	NA	Hypotension Electrolyte imbalance Renal insufficiency
MRA	NA	NA	RCT[38,a]	EO[39-41]	Hyperkalemia Renal insufficiency
Sacubitril/valsartan	NA	NA	MCS[42-46,b]	NA	Hypotension Electrolyte imbalance Renal insufficiency
SGLT-2 inhibitors	NA	NA	MCS[42,47-50,b]	MCS[51-55,b]	Hypoglycemia Infection Kidney failure
GLP-1 agonists	NA	NA	NA	NA	Gastrointestinal symptoms Hypoglycemia Dehydration Pancreatitis
Vericiguat	NA	NA	NA	NA	NRU[c]
Digoxin	NA	NA	NA	NA	NRU[c]
Ivabradine	NA	NA	NA	NA	NRU[c]
Hydralazine	NA	NA	NA	NA	NRU[c]
Isosorbide dinitrate	NA	NA	NA	RCT[56,a]	Hypotension, headache

Abbreviations: ACE, angiotensin-converting enzyme; ACHD, adult congenital heart disease; ARB, angiotensin II receptor blocker; EO, expert opinion; GLP-1, glucagon-like peptide-1; MCS, multiple cohort studies; MRA, mineralocorticoid receptor antagonist; NA, not available; NRU, no reported use; RCT, randomized clinical trial; SGLT-2, sodium-glucose cotransporter 2; sRV, systemic right ventricle; SV, single ventricle; TGA, transposition of the great arteries; TOF, tetralogy of Fallot.

[a] Randomized clinical trial supporting use.

[b] Multiple cohort studies supporting use.

[c] No reported use in ACHD.

of adult survivors with TOF (although in only 6% with LV ejection fraction <45%) and was associated with increased mortality (30% additional risk of death or sustained ventricular tachycardia for every 5% decrease in LV ejection fraction).[30]

Among patients with surgically repaired TOF, moderate or greater tricuspid regurgitation develops in 20% to 32%, atrial tachycardia and atrial fibrillation in approximately 20% to 45% by age 45 years,[7] and prosthetic pulmonary valve dysfunction (stenosis, regurgitation, or both) in 25% to 40%.[28] Additionally, patients with surgically repaired TOF are at increased risk of infective endocarditis (10-year risk, 1.3%-1.6%)[29] and ventricular arrythmias, most commonly monomorphic ventricular tachycardia, which affect 15% of adult patients with surgically repaired TOF.[31] In a multicenter cohort study of 1147 patients with repaired TOF, 8% of patients (mean age, 27 years at enrollment [SD, 14]) experienced a composite outcome of all-cause mortality, resuscitated cardiac arrest, or sustained ventricular tachycardia after a median follow-up of 8 years (IQR, 3.8-12.7 years).[32] Risk stratification for sudden cardiac death is challenging and should be performed by cardiologists experienced in adult CHD (ACHD) care (Table 1).[26]

Management of Complications

Among patients with repaired TOF who have severe pulmonary regurgitation and severe heart failure symptoms, a propensity-matched cohort study (1143 patients) showed that pulmonary valve replacement compared with no replacement was associated with improved symptoms and survival (10-year freedom from death or sustained ventricular tachycardia, 92% vs 85%; hazard ratio for death and sustained ventricular tachycardia, 0.41; 95% CI, 0.21-0.81).[32] Surgical pulmonary valve replacement is a low-risk procedure (<0.5%-1.0% perioperative mortality) in adults with repaired TOF.[33] For some patients, the pulmonary valve can be replaced through an interventional cardiology procedure instead of surgery.[14,33,34]

Patients with surgically repaired TOF who develop heart failure should receive guideline-directed medical therapy management used in the care of adults with acquired heart failure with reduced ejection fraction (Table 2).[14,35-56] More than 90% of monomorphic ventricular tachycardias occur due to macroreentry caused by slow conduction through scar tissue, which makes them amenable to catheter ablation.[14,23,57,58]

TGA

Anatomy and Early Management

In TGA, the aorta arises from the RV and the pulmonary artery arises from the LV, so RV deoxygenated blood is recirculated systemically and LV oxygenated blood is recirculated to the lung (Figure 1B), causing severe hypoxia with cyanosis and acidosis after birth. Infants with TGA may also have additional congenital anomalies, including malalignment ventricular septal defect, subpulmonary stenosis, coarctation of the aorta, and anomalies of atrial situs. In 90% to 95% of patients, the right atrium connects to the RV (termed *D-loop TGA*), which results in ineffective parallel circulations in which deoxygenated blood remains in the systemic circulation and oxygenated blood remains in the pulmonary circulation (Figure 1B); initial palliative neonatal management includes establishing a wide open atrial communication via a transcatheter balloon septostomy to allow mixing of oxygenated and deoxygenated blood, which creates some degree of systemic arterial flow with oxygenated blood.[59,60]

Arterial Switch Operations

Current surgical repair of D-loop TGA is typically a so-called arterial switch operation (Figure 3B), which restores appropriate anatomic connection between the LV and aorta and the RV and pulmonary artery. The arterial switch operation is accomplished by transecting the great arteries above their roots, transforming the native LV-pulmonary valve into an LV-neoaortic valve and neoaortic root, and transforming the native RV-aortic valve into an RV-neopulmonary valve and neopulmonary root. In addition, the coronary arteries are transferred from the native aortic root to the neoaorta so that they carry oxygenated blood to the heart (Figure 3B). Patients with D-loop TGA who undergo arterial switch have a 93% to 97% 30-year survival, 75% to 77% freedom from reintervention at 25 years, and New York Heart Association (NYHA) class I (asymptomatic) functional status of 98% at 25 years.[14]

Long-Term Complications After Arterial Switch Operation

Long-term complications of arterial switch surgery include narrowing or kinking at the sites where the pulmonary artery and aorta were switched, leading to supravalvular pulmonary or aortic stenosis (supravalvular pulmonary stenosis is noted by echocardiography in approximately 80% of patients at 35 years, although moderate or greater supravalvular aortic stenosis occurs much less frequently [3.2% at 18-19 years]). Aortic root dilation occurred in 66% to 75% of 124 patients with D-loop TGA assessed by serial transthoracic echocardiograms up to 25 years after arterial switch, with moderate or greater valve regurgitation reported in 11% to 14% of patients.[61] Asymptomatic myocardial ischemia, evidenced by exercise-induced myocardial perfusion defects, may occur in up to 10% of patients by age 20 years due to kinking, narrowing, and excessive tortuosity of reimplanted coronary arteries.

Management of Late Complications After Arterial Switch Operations

By age 35 years, approximately 13% of patients who underwent arterial switch surgery require aortic valve or aortic root surgery to repair the aortic root dilation.[62] Coronary intervention occurs most commonly in youth to correct coronary artery stenosis or obstruction at the site of reimplantation.[4]

Atrial Switch Operations

Before the era of arterial switch operation, which was introduced in 1976, D-loop TGA surgical repair was performed with a venous return redirection surgery (atrial switch operations) that funnels pulmonary venous return to the tricuspid valve and caval return to the mitral valve by surgically created intracardiac pathways (Figure 3B). Although these procedures allowed deoxygenated and oxygenated blood to flow to appropriate circulations, the RV directly ejects blood into the systemic circulation, which increases the risk of progressive RV failure and regurgitation of the tricuspid valve that is serving as the main valve for systemic circulation.[63,64] Although the atrial switch procedure is no longer routinely performed, there are long-term survivors of this repair.

Long-Term Complications After Atrial Switch Operation

Patients with D-loop TGA who underwent atrial switch operations have RV failure rates of 30% to 50% at 25 years after atrial switch repair and a combined event rate of death, mechanical circulatory support, and heart transplant of 8.8/1000 person-years.[65,66] Atrial tachycardia develops in 48% to 63% of patients at 32 to 40 years postsurgery and sinus node dysfunction occurs in 50% to 65% of patients at 35 years postsurgery, and these rates increase with age.[66] Sudden cardiac death occurs in up to 15% of patients with D-loop TGA after atrial switch at a mean age of 32.6 years (SD, 6.4 years),[66,67] is usually associated with polymorphic ventricular tachycardia or ventricular fibrillation, and is frequently triggered by exercise (Table 1).

Management of Late Complications After Atrial Switch Operations

Studies of patients with RV failure after atrial switch have demonstrated improvements in N-terminal pro–brain natriuretic peptide levels, systemic RV function exercise capacity, and quality of life with use of sacubitril/valsartan.[14,42-45,68] Although long-term data are lacking, in observational cohorts, sodium-glucose cotransporter 2 inhibitors are associated with improvements in NYHA class and 6-minute walking test distance, echocardiographic findings (RV global longitudinal strain decrease, −1.4% point per month, $P < .001$; and fractional area increase, 3.2% point per month, $P = .002$), and reduced N-terminal pro–brain natriuretic peptide level (Table 2) and were well tolerated.[47-49]

Patients with end-stage RV failure (presenting with NYHA III/IV functional limitations, recurrent hospital admissions, and low cardiac output state with cardiorenal syndrome) should be referred for heart transplant evaluation, although pulmonary hypertension identified by right-sided heart catheterization may preclude heart transplant.[66,69] Atrial tachycardia can be treated with catheter ablation.[70] Patients with ventricular tachycardia after D-loop TGA atrial switch may undergo implantable cardioverter defibrillator placement and be treated with antiarrhythmic pharmacotherapy, whereas those with heart failure may be treated with guideline-directed medical therapy (recognizing the lack of data supporting effect on mortality), evaluation for heart transplant, or both.[16]

In a subset of patients with D-loop TGA after atrial switch, severe atrioventricular valve regurgitation may accelerate heart failure progression and symptoms (hazard ratio for death, transplant, and mechanical circulatory support, 2.08; 95% CI, 0.90-4.79).[66] Surgical valve replacement is usually associated with substantial procedural risks; outcomes appear worse after development of moderate to severe RV dysfunction.[71] Recently, transcatheter edge-to-edge valve repair has been performed in select patients in experienced centers, with reasonable short-term results.[72]

Single Ventricle With Fontan Circulation

Anatomy and Early Management

The term *single ventricle* encompasses a wide range of congenital anomalies in which either a single or combined ventricular chamber pumps to both pulmonary and systemic arterial circulations. The most common congenital lesions presenting as single ventricle are hypoplastic left heart syndrome, tricuspid atresia, and double inlet ventricles (Figure 1C).[73] Cyanosis due to intracardiac venous return mixing occurs in each of these uncorrected lesions.

Surgical plans for this group of patients most often follows a standardized and staged approach that is palliative to improve survival and symptoms because the anatomy cannot be corrected to a completely normal state.[74] Neonates whose single ventricle exits toward the typically low-resistance pulmonary vascular bed have excess flow and pressure in the lung, which leads to development of pulmonary vascular remodeling and pulmonary vascular disease. For these neonates, creation of a mechanical flow barrier, or pulmonary arterial band, may restrict pulmonary blood flow and balance circulation between the pulmonary and systemic circulation, protecting from the effect of pulmonary overcirculation (Figure 3C). Neonates with overly restricted pulmonary blood flow resulting in severe cyanosis may be treated with the creation or enlargement of flow pathways to the lung by stenting of the ductus arteriosus to keep it open as a source of pulmonary blood flow, or they may be treated with creation of a surgical connection between the subclavian artery and the pulmonary artery (ie, Blalock-Taussig-Thomas shunt or Norwood procedure in patients with hypoplastic left heart syndrome) (Figure 3C).[74] Neonates with single ventricle, without severe cyanosis, and with protective pressure gradient across the pulmonary valve may not require any early intervention. At age 6 to 9 months, infants with single ventricle are usually considered for direct surgical anastomosis of the superior vena cava to the right pulmonary artery (Figure 3C) (bidirectional cavopulmonary anastomosis, often referred to as Glenn surgery) to reduce ventricular volume overload and improve oxygen saturation. The final step in management of single-ventricle physiology, typically performed between age 18 to 36 months, involves complete surgical separation of systemic and pulmonary venous return, leading to often near-normalization of systemic oxygen saturation. Pioneered by Fontan and Baudet,[75] a direct connection of the inferior vena cava to the pulmonary artery is generally created with an intra-atrial lateral tunnel (baffle) or extracardiac tube (conduit), allowing redirection of systemic venous return to the pulmonary circulation without an interposing ventricular pumping chamber, referred to as Fontan circulation (Figure 3C).

Long-Term Complications

Patients with Fontan circulation experience substantial cardiovascular morbidity and mortality, including thromboembolic events, arrhythmias, and potential for relatively unusual heart failure phenotypes, such as lymphatic complications: protein-losing enteropathy (enteral loss of plasma proteins), plastic bronchitis (thick rubbery proteinaceous casts in the airways, in part caused by severe lymphatic obstruction in the lungs), progressive cyanosis, and refractory ascites with reduced or preserved ventricular function (collectively, risk for Fontan circulatory failure in this group approximates 40% after 35 years of follow-up of surgery survivors).[76,77]

In a large registry of 1006 patients who survived Fontan surgery and were followed up to 25 years, 31 underwent modernization (conversion) Fontan surgery largely to address refractory arrhythmias due to severely dilated atrial chamber, and 7 underwent Fontan takedown surgery (surgical reconnection of the inferior vena cava to the atrium) because of very high central venous pressure and low cardiac output; 11 patients progressed to NYHA functional class III/IV, 15 developed protein-losing enteropathy/plastic bronchitis, 16 underwent heart transplant, and 42 died.[78] Among these registry patients, freedom from major adverse events was 83% (95% CI, 79%-86%) at 15 years, 70% (95% CI, 63%-76%) at 20 years, and 56% (95% CI, 44%-66%) at 25 years after Fontan surgery; these patients may develop NYHA functional class III or IV (0.35% per person-year) and have increased risk of early death or heart transplant requirement (0.36% per person-year).[78,79] Fontan circulatory failure (1.5%/person-year) is defined broadly as a dysfunction of the Fontan circulation that affects a patient's ability to carry out daily life activities.[79,80] Fontan circulation has limited reserve from key compensatory mechanisms (pulmonary vascular recruitment, splanchnic modulation, and systemic venoreactivity). It is characterized by a preload-limited systemic ventricle and obligate systemic venous hypertension that can contribute to lymphatic complications.[76] There are 2 predominant clinical phenotypes of Fontan circulatory failure: (1) primary ventricular failure, which is less common and typically managed with conventional heart failure treatment algorithms; and (2) preserved ventricular function,[76] which is associated with lymphatic complications or refractory ascites.[81]

Chronically elevated central venous pressure, low cardiac output, and hypoxemia in patients with Fontan circulation impair oxygen delivery to centrilobar hepatocytes, triggering fibrotic changes leading to Fontan-associated liver disease, which can range from asymptomatic hepatic fibrosis to advanced cirrhosis.[82] At 10 years after Fontan surgery, approximately 40% of patients have bridging hepatic fibrosis at liver biopsy.[83] Hypervascular hepatic nodules occur in 20% to 30% of patients with Fontan circulation, including focal nodular hyperplasia-like lesions and large regenerative nodules with risk for progression to hepatocellular carcinoma.[84] As standardization of liver assessment strategies for Fontan-associated liver disease monitoring evolves, proactive surveillance (including measurement of routine blood liver function tests, abdominal ultrasonographic and elastographic imaging studies, and oncologic surveillance with serum alpha fetoprotein) is important for all Fontan surgery survivors because of the universal and progressive nature of Fontan-associated liver disease.[82]

Management of Late Complications

Medical therapy for Fontan circulatory failure is largely extrapolated from evidence in adults with noncongenital biventricular heart failure (Table 2) and remains an area of high uncertainty.[14] Although to our knowledge there are no published trials comparing use of antiplatelet therapy vs anticoagulation to reduce thrombotic risk in Fontan survivors, use of either therapy is associated with lower risk of thrombosis.[85] In a crossover trial of 26 patients with a previous Fontan operation, carvedilol was not associated with improved exercise performance and was associated with mildly increased N-terminal pro–brain natriuretic peptide levels.[86] Mineralocorticoid receptor antagonists, particularly high-dose spironolactone, may improve heart failure symptoms in specific subgroups (eg, patients with hepatic congestion or protein-losing enteropathy) by reducing protein enteric loss, although data are limited (Table 2).[39,87]

Recent studies suggest that sodium-glucose cotransporter 2 inhibitors may be safe and well tolerated, although definitive improvements in functional parameters were not statistically significant.[51-55] Because of conflicting evidence on surrogate end points, pulmonary vasodilators (eg, sildenafil) are not recommended for Fontan survivors.[88]

For patients with Fontan circulatory failure, severe atrioventricular valve regurgitation may accelerate heart failure progression and symptoms; for these patients, the approach to therapy and assessment of outcomes are similar to that discussed for adults with D-loop TGA. Adults who undergo atriopulmonary Fontan procedure often develop atrial tachycardia or atrial fibrillation if they survive into their third or fourth decade; although less common with the lateral tunnel and extracardiac conduit types (15-year risk of atrial arrhythmia after initial Fontan procedure, 17% and 8%, respectively), atrial fibrillation is reported in approximately 10% of Fontan survivors and incidence of atrial arrhythmia markedly increases with age.[89-91]

Similar to atrial switch anatomies, catheter ablation often requires complex access to the pulmonary venous atrium, where most atrial tachycardia substrates (both intra-atrial reentrant tachycardia and atrial fibrillation targets) reside in patients after Fontan surgery.[92] Ventricular tachycardia and sudden cardiac death (reported in up to 5% of Fontan survivors) are less common among patients who undergo Fontan procedure compared with those with TOF and systemic RV. However, ventricular tachycardia and sudden cardiac death can occur, typically after development of advanced Fontan circulatory failure (Table 1).[20,93]

Arrhythmias | A few generalizations regarding arrhythmia management in the ACHD population follow. Essentially all patients with ACHD are at heightened risk for atrial arrhythmias, most often typical cavotricuspid isthmus–dependent flutter and atypical atriotomy-based flutters, collectively referred to as atrial tachycardia. Many patients with ACHD have indications for chronic pacing due to sinus node dysfunction or atrioventricular block, and thus most with indications for implantable cardioverter defibrillator currently undergo transvenous implantation, which allows for antibradycardia pacing, antitachycardia pacing, and defibrillation (although future advances in leadless pacemaker technology may challenge this practice). Patients with complex intracardiac or extracardiac surgically constructed pathways for blood flow (respectively named baffles and conduits) are at risk for transvenous lead-related baffle pathway obstruction and paradoxical emboli through intracardiac baffle leaks. Select patients without pacing indications may be candidates for the subcutaneous implantable cardioverter defibrillator, which avoids lead-related risk, yet many patients with ACHD are less favorable candidates due to (1) bradycardia pacing indications that may not be adequately cared for by leadless pacing systems; (2) baseline QRS abnormalities that preclude effective sensing; or (3) both. Certain surgical repairs (eg, Fontan surgery; see below) preclude systemic venous access to a subpulmonic ventricle, making routine transvenous implantable cardioverter defibrillator less possible, and a high proportion of patients undergo epicardial implantation, which may require a repeat sternotomy (with its associated morbidity).[14] Risk factors for ventricular tachycardia and sudden cardiac death are listed for each lesion in Table 1.

Atrial Fibrillation One in 10 patients with ACHD will develop atrial fibrillation by age 40 years and 10% of patients develop atrial fibrillation within 1 year of atrial tachycardia ablation.[94,95] Atrial fibrillation in ACHD is almost always persistent (ie, not self-resolving paroxysms); typically develops in the context of substantial atrial dilation, chronic pressure/volume overload, cardiac fibrosis, or all 3; and tends to be associated with symptoms of lower functional capacity. Conversion to sinus rhythm and use of oral anticoagulation (vitamin K antagonists or direct oral anticoagulants, based on lesion complexity and patient comorbidities) are recommended in current care guidelines.[14]

Cardiac Resynchronization Therapy Cardiac resynchronization therapy represents a pacing strategy that seeks to restore synchronous contraction to a ventricle to improve its function and reduce heart failure symptoms.[96] Studies have shown variable improvements in heart failure symptoms and mortality associated with cardiac resynchronization therapy in children and adults with CHD, likely due to the heterogeneity in intraventricular conduction delay patterns (left vs right bundle-branch block), systemic ventricular morphology (RV vs LV vs single-ventricle Fontan circulation), and pacing implantation method (transvenous vs epicardial).[97-100] In attempts to mimic native ventricular activation, conduction system pacing techniques can lead to synchronous ventricular activation with improvement in symptoms, using a single ventricular lead.[101]

Heart Transplant and Mechanical Circulatory Support | Patients with end-stage heart failure due to CHD may undergo heart transplant, which is definitive therapy.[102] The International Society for Heart and Lung Transplantation reported that from 1992 to 2003, 1.9% of transplant recipients were patients with ACHD, a proportion that increased to 3.1% from 2009 to 2017.[103,104] Similarly, the United Network for Organ Sharing reported that ACHD accounted for 1.8% of heart transplants from 1992 to 2001 and 2.5% from 2002 to 2009, with a higher percentage increase in younger patients (18-39 years).[103] However, access to heart transplant is limited for multiple factors, including (1) lower priority in organ allocation owing to appearing less critically ill by standard criteria, such as the need for inotropic support, compared with other candidates for heart transplant; (2) high rates of allosensitization from multiple blood transfusions and implanted homograft tissues; (3) anatomic factors; and (4) variable risk models and poorly defined criteria for the timing of heart transplant listing. One-year survival after heart transplant is lower for patients with ACHD compared with those without ACHD (1-year survival for ACHD vs nonischemic cardiomyopathy, 80% vs 85%; hazard ratio, 2.21; 95% CI, 1.62-3.02; $P < .001$).[102,104] However, for patients with ACHD who survive the first year after transplant, the overall survival is not inferior (median survival in ACHD vs nonischemic cardiomyopathy, 20.1 vs 14.4 years; $P < .05$).[102,104]

The International Society for Heart and Lung Transplantation reported that risk factors for early mortality after heart transplant include graft failure (41%), multiple organ failure (14%), infection (13%), surgical complications (7%), and cerebrovascular events (7%).[102,103,105]

Mechanical circulatory support (such as ventricular assist device) is increasingly used for patients with ACHD with end-stage heart failure as bridge to heart transplant, bridge to candidacy, or destination therapy. The Scientific Registry of Transplant Recipients reported that mechanical circulatory support increased from less than 4% (1993-1996) to 14% (2010-2012).[102,106] However, use of mechanical circulatory support may be limited by specific risk factors in patients with ACHD, including (1) smaller body size; (2) difficulty positioning cannula because of anatomic constraints, such as prominent RV trabeculae/moderator band and subvalvular apparatus in systemic RV; (3) prior sternotomies; (4) presence of previous shunt repairs with multiple suture lines in patients with Fontan circulation; (5) need for subpulmonic mechanical inotropic support; (6) postoperative volume overload; and (7) postoperative acute kidney and liver injury. The Interagency Registry for Mechanically Assisted Circulatory Support reported that among 128 patients with ACHD and mechanical circulatory support, 1-year survival was 52%, transplant rate was 20%, and mortality rate was 27%, similar to the rates in 512 matched patients without ACHD.[102,107]

Practical Considerations

Survivors of cyanotic CHD experience challenges due to many factors, including advanced cardiac care during infancy (including neonatal or early need for cardiopulmonary bypass), prolonged and multiple hospitalizations, cyanosis, early exposure to pharmacotherapy, and limitations on life experiences.[108,109] Among 2192 US adolescents (11-17 years) with prevalent CHD identified by diagnosis code using electronic health records of multiple health care databases and linked to a pertinent diagnostic code for mental illness, up to 20% experienced anxiety disorders, attention disorders, conduct disorders, behavior/impulse control disorders, and mood disorders.[110] In a cross-sectional survey study, 3815 patients with ACHD from 15 countries (mean [SD] age, 34.8 [12.9] years; 52.7% female) completed the Hospital Anxiety and Depression Scale (HADS), which includes subscales for anxiety (HADS-A) and depression (HADS-D).[111] In this study, 18.3% of patients had symptoms of anxiety according to an elevated HADS-A score, 2.9% had depression according to an elevated HADS-D score, and 8.9% had both anxiety and depression according to elevated HADS-A and HADS-D scores.[111] Quality of life and health status decreased with increasing HADS-A and HADS-D scores ($P < .001$).[111]

Women with repaired cyanotic CHD are increasingly entering child-bearing age, and family planning, preconceptional counseling, and pregnancy surveillance are mandatory and require expert and multidisciplinary care.[112]

Due to the complex congenital and surgically modified anatomic characteristics of cyanotic CHD, safe and effective management for adult survivors of cyanotic CHD typically requires specialized expertise, advanced imaging integration, and procedural experience usually available only at dedicated ACHD referral centers. There are 60 accredited ACHD centers in the United States. Because adult survivors of cyanotic CHD are at increased risk of kidney disease; hematologic disorders, such as secondary erythrocytosis; liver disease (cirrhosis); and cancer, appropriate care requires effective integration of primary care physicians, internists, psychologists, and ACHD care clinicians to ensure lifelong surveillance for multisystemic disorders.[113]

Limitations

This review has several limitations. First, some relevant articles may have been missed. Second, we did not perform a systematic evaluation of the quality of reviewed evidence. Third, care of patients with cyanotic CHD varies according to regional and logistic factors and local expert opinion.

Conclusions

With surgical intervention, survival to adulthood is common among patients with TOF, D-loop TGA, and single ventricle. However, these survivors of cyanotic CHD are at risk of premature death and of developing valve dysfunction, arrhythmias, and heart failure. Optimal care involves multidisciplinary management, including pediatric and adult cardiologists, congenital cardiac surgeons, and electrophysiologists.

ARTICLE INFORMATION

Accepted for Publication: May 4, 2026.

Published Online: June 8, 2026. doi:10.1001/jama.2026.8806

Author Affiliations: Adult Congenital Heart Disease Program, Department of Cardiovascular and Thoracic Medicine, IRCCS Azienda Ospedaliero-Universitaria di Bologna, Bologna, Italy (Egidy Assenza); Department of Cardiology, Boston Children's Hospital, Boston, Massachusetts (O'Leary, Landzberg); Harvard Medical School, Boston, Massachusetts (O'Leary, Kwon, Landzberg); Adult Congenital Heart Centre and National Centre for Pulmonary Hypertension, Royal Brompton & Harefield Hospitals, Guys and St Thomas' NHS Trust, London, United Kingdom (Brida); National Heart and Lung Institute, Imperial College London, London, United Kingdom (Brida); Department of Medical Rehabilitation, Medical Faculty University of Rijeka, Rijeka, Croatia (Brida); Department of Cardiac Surgery, Boston Children's Hospital, Boston, Massachusetts (Kwon).

Conflict of Interest Disclosures: Dr O'Leary reported educational consulting fees from Johnson & Johnson MedTech and scientific advisory board fees from EnChannel Medical outside the submitted work. No other disclosures were reported.

Submissions: We encourage authors to submit papers for consideration as a Review. Please contact Kristin Walter, MD, at kristin.walter@jamanetwork.org.

REFERENCES

1. Liu Y, Chen S, Zühlke L, et al. Global birth prevalence of congenital heart defects 1970-2017: updated systematic review and meta-analysis of 260 studies. *Int J Epidemiol*. 2019;48(2):455-463. doi:10.1093/ije/dyz009

2. Sommer RJ, Hijazi ZM, Rhodes JF. Pathophysiology of congenital heart disease in the adult, part III: complex congenital heart disease. *Circulation*. 2008;117(10):1340-1350. doi:10.1161/CIRCULATIONAHA.107.714428

3. Nollert G, Fischlein T, Bouterwek S, Böhmer C, Klinner W, Reichart B. Long-term survival in patients with repair of tetralogy of Fallot: 36-year follow-up of 490 survivors of the first year after surgical repair. *J Am Coll Cardiol*. 1997;30(5):1374-1383. doi:10.1016/S0735-1097(97)00318-5

4. Engele LJ, van der Palen RLF, Joosen RS, et al. Clinical course of TGA after arterial switch operation in the current era. *JACC Adv*. 2023;3(2):100772. doi:10.1016/j.jacadv.2023.100772

5. Venkatesh P, Evans AT, Maw AM, et al. Predictors of late mortality in D-transposition of the great arteries after atrial switch repair: systematic review and meta-analysis. *J Am Heart Assoc*. 2019;8(21):e012932. doi:10.1161/JAHA.119.012932

6. Moons P, Bovijn L, Budts W, Belmans A, Gewillig M. Temporal trends in survival to adulthood among patients born with congenital heart disease from 1970 to 1992 in Belgium. *Circulation*. 2010;122(22):2264-2272. doi:10.1161/CIRCULATIONAHA.110.946343

7. Smith CA, McCracken C, Thomas AS, et al. Long-term outcomes of tetralogy of Fallot: a study from the Pediatric Cardiac Care Consortium. *JAMA Cardiol*. 2019;4(1):34-41. doi:10.1001/jamacardio.2018.4255

8. Kauling RM, Ünlütürk S, Cuypers JAAE, et al. Long term outcome after surgical tetralogy of Fallot repair at young age: longitudinal follow-up up to 50 years after surgery. *Int J Cardiol*. 2025;423:133005. doi:10.1016/j.ijcard.2025.133005

9. Couperus LE, Vliegen HW, Zandstra TE, et al. Long-term outcome after atrial correction for transposition of the great arteries. *Heart*. 2019;105(10):790-796. doi:10.1136/heartjnl-2018-313647

10. Fricke TA, Buratto E, Weintraub RG, et al. Long-term outcomes of the arterial switch operation. *J Thorac Cardiovasc Surg*. 2022;163(1):212-219. doi:10.1016/j.jtcvs.2021.01.134

11. Dennis M, Zannino D, du Plessis K, et al. Clinical outcomes in adolescents and adults after the Fontan procedure. *J Am Coll Cardiol*. 2018;71(9):1009-1017. doi:10.1016/j.jacc.2017.12.054

12. Egidy Assenza G, Krieger EV, Baumgartner H, et al. AHA/ACC vs ESC guidelines for management of adults with congenital heart disease: JACC guideline comparison. *J Am Coll Cardiol*. 2021;78(19):1904-1918. doi:10.1016/j.jacc.2021.09.010

13. Mylotte D, Pilote L, Ionescu-Ittu R, et al. Specialized adult congenital heart disease care: the impact of policy on mortality. *Circulation*. 2014;129(18):1804-1812. doi:10.1161/CIRCULATIONAHA.113.005817

14. Gurvitz M, Krieger EV, Fuller S, et al; Writing Committee Members. 2025 ACC/AHA/HRS/ISACHD/SCAI guideline for the management of adults with congenital heart disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2026;153(8):e115-e251. doi:10.1161/CIR.0000000000001402

15. Perloff JK, Rosove MH, Child JS, Wright GB. Adults with cyanotic congenital heart disease: hematologic management. *Ann Intern Med*. 1988;109(5):406-413. doi:10.7326/0003-4819-109-5-406

16. Ladouceur M, Van De Bruaene A, Kauling R, et al. A new score for life-threatening ventricular arrhythmias and sudden cardiac death in adults with transposition of the great arteries and a systemic right ventricle. *Eur Heart J*. 2022;43(28):2685-2694. doi:10.1093/eurheartj/ehac288

17. Stephenson EA, Lu M, Berul CI, et al; Pediatric Heart Network Investigators. Arrhythmias in a contemporary Fontan cohort: prevalence and clinical associations in a multicenter cross-sectional study. *J Am Coll Cardiol*. 2010;56(11):890-896. doi:10.1016/j.jacc.2010.03.079

18. Pujol C, Hessling G, Telishevska M, et al. Prevalence and treatment outcomes of arrhythmias in patients with single ventricle physiology over the age of 40 years. *J Clin Med*. 2022;11(21):6568. doi:10.3390/jcm11216568

19. Weipert J, Noebauer C, Schreiber C, et al. Occurrence and management of atrial arrhythmia after long-term Fontan circulation. *J Thorac Cardiovasc Surg*. 2004;127(2):457-464. doi:10.1016/j.jtcvs.2003.08.054

20. Giacone HM, Chubb H, Dubin AM, et al. Outcomes after development of ventricular arrhythmias in single ventricular heart disease patients with Fontan palliation. *Circ Arrhythm Electrophysiol*. 2023;16(6):e011143. doi:10.1161/CIRCEP.122.011143

21. Ghonim S, Gatzoulis MA, Ernst S, et al. Predicting survival in repaired tetralogy of Fallot: a lesion-specific and personalized approach. *JACC Cardiovasc Imaging*. 2022;15(2):257-268. doi:10.1016/j.jcmg.2021.07.026

22. Atallah J, Gonzalez Corcia MC, Walsh EP; Participating Members of the Pediatric and Congenital Electrophysiology Society. Ventricular arrhythmia and life-threatening events in patients with repaired tetralogy of Fallot. *Am J Cardiol*. 2020;132:126-132. doi:10.1016/j.amjcard.2020.07.012

23. Kapel GF, Sacher F, Dekkers OM, et al. Arrhythmogenic anatomical isthmuses identified by electroanatomical mapping are the substrate for ventricular tachycardia in repaired tetralogy of Fallot. *Eur Heart J*. 2017;38(4):268-276. doi:10.1093/eurheartj/ehw202

24. Wald RM, Valente AM, Marelli A. Heart failure in adult congenital heart disease: emerging concepts with a focus on tetralogy of Fallot. *Trends Cardiovasc Med*. 2015;25(5):422-432. doi:10.1016/j.tcm.2014.11.011

25. Arnaert S, De Meester P, Troost E, et al. Heart failure related to adult congenital heart disease: prevalence, outcome and risk factors. *ESC Heart Fail*. 2021;8(4):2940-2950. doi:10.1002/ehf2.13378

26. Baumgartner H, De Backer J, Babu-Narayan SV, et al; ESC Scientific Document Group. 2020 ESC guidelines for the management of adult congenital heart disease. *Eur Heart J*. 2021;42(6):563-645. doi:10.1093/eurheartj/ehaa554

27. Apitz C, Webb GD, Redington AN. Tetralogy of Fallot. *Lancet*. 2009;374(9699):1462-1471. doi:10.1016/S0140-6736(09)60657-7

28. Kwon MH, Bacha EA. Pulmonary valve-sparing techniques for tetralogy of Fallot: a systematic approach for maximizing success and minimizing risk. *Semin Thorac Cardiovasc Surg Pediatr Card Surg Annu*. 2020;23:24-28. doi:10.1053/j.pcsu.2020.02.005

29. Geva T, Wald RM, Bucholz E, et al; American Heart Association Council on Lifelong Congenital Heart Disease and Heart Health in the Young; Council on Cardiovascular Surgery and Anesthesia; Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing. Long-term management of right ventricular outflow tract dysfunction in repaired tetralogy of Fallot: a scientific statement from the American Heart Association. *Circulation*. 2024;150(25):e689-e707. doi:10.1161/CIR.0000000000001291

30. Egbe AC, Adigun R, Anand V, et al. Left ventricular systolic dysfunction and cardiovascular outcomes in tetralogy of Fallot: systematic review and meta-analysis. *Can J Cardiol*. 2019;35(12):1784-1790. doi:10.1016/j.cjca.2019.07.634

31. Khairy P, Aboulhosn J, Gurvitz MZ, et al; Alliance for Adult Research in Congenital Cardiology (AARCC). Arrhythmia burden in adults with surgically repaired tetralogy of Fallot: a multi-institutional study. *Circulation*. 2010;122(9):868-875. doi:10.1161/CIRCULATIONAHA.109.928481

32. Bokma JP, Geva T, Sleeper LA, et al. Improved outcomes after pulmonary valve replacement in repaired tetralogy of Fallot. *J Am Coll Cardiol*. 2023;81(21):2075-2085. doi:10.1016/j.jacc.2023.02.052

33. Van den Eynde J, Sá MPBO, Vervoort D, et al. Pulmonary valve replacement in tetralogy of Fallot: an updated meta-analysis. *Ann Thorac Surg*. 2022;113(3):1036-1046. doi:10.1016/j.athoracsur.2020.11.040

34. Patel ND, Levi DS, Cheatham JP, Qureshi SA, Shahanavaz S, Zahn EM. Transcatheter pulmonary valve replacement: a review of current valve technologies. *J Soc Cardiovasc Angiogr Interv*. 2022;1(6):100452. doi:10.1016/j.jscai.2022.100452

35. Misra A, Rodriguez-Monserrate CP, Gauvreau K, et al. Propensity score analysis of possible medication effects on outcomes in patients with systemic right ventricles. *JACC Adv*. 2024;4(1):101443. doi:10.1016/j.jacadv.2024.101443

36. Babu-Narayan SV, Uebing A, Davlouros PA, et al. Randomised trial of ramipril in repaired tetralogy of Fallot and pulmonary regurgitation: the APPROPRIATE study (Ace Inhibitors for Potential Prevention of the Deleterious Effects of Pulmonary Regurgitation in Adults With Repaired Tetralogy of Fallot). *Int J Cardiol*. 2012;154(3):299-305. doi:10.1016/j.ijcard.2010.09.057

37. van der Bom T, Winter MM, Bouma BJ, et al. Effect of valsartan on systemic right ventricular function: a double-blind, randomized, placebo-controlled pilot trial. *Circulation*. 2013;127(3):322-330. doi:10.1161/CIRCULATIONAHA.112.135392

38. Dos L, Pujadas S, Estruch M, et al. Eplerenone in systemic right ventricle: double blind randomized clinical trial: the EVEDES study. *Int J Cardiol*. 2013;168(6):5167-5173. doi:10.1016/j.ijcard.2013.07.163

39. Ringel RE, Peddy SB. Effect of high-dose spironolactone on protein-losing enteropathy in patients with Fontan palliation of complex congenital heart disease. *Am J Cardiol*. 2003;91(8):1031-1032, A1039. doi:10.1016/s0002-9149(03)00135-8

40. Okada S, Muneuchi J, Nagatomo Y, et al. Successful treatment of protein-losing enteropathy and plastic bronchitis by biphasic cuirass ventilation in a patient with failing Fontan circulation. *Int Heart J*. 2018;59(4):873-876. doi:10.1536/ihj.17-292

41. Grattan MJ, McCrindle BW. Recurrent exacerbations of protein-losing enteropathy after initiation of growth hormone therapy in a Fontan patient controlled with spironolactone. *Congenit Heart Dis*. 2010;5(2):165-167. doi:10.1111/j.1747-0803.2009.00320.x

42. Neijenhuis RML, Nederend M, van Groningen AE, et al. Sacubitril/valsartan is associated with improvements in quality of life in adult congenital heart disease patients with systemic right ventricular failure. *Open Heart*. 2025;12(1):e003009. doi:10.1136/openhrt-2024-003009

43. Fusco F, Scognamiglio G, Merola A, et al. Safety and efficacy of sacubitril/valsartan in patients with a failing systemic right ventricle: a prospective single-center study. *Circ Heart Fail*. 2023;16(2): e009848. doi:10.1161/CIRCHEARTFAILURE.122.009848

44. Nederend M, Kiès P, Regeer MV, et al. Tolerability and beneficial effects of sacubitril/valsartan on systemic right ventricular failure. *Heart*. 2023;109(20):1525-1532. doi:10.1136/heartjnl-2022-322332

45. Zandstra TE, Nederend M, Jongbloed MRM, et al. Sacubitril/valsartan in the treatment of systemic right ventricular failure. *Heart*. 2021;107(21):1725-1730. doi:10.1136/heartjnl-2020-318074

46. Ciriello GD, Altobelli I, Fusco F, et al. Sacubitril/valsartan and dapagliflozin in patients with a failing systemic right ventricle: effects on the arrhythmic burden. *J Clin Med*. 2024;13(24):7659. doi:10.3390/jcm13247659

47. Neijenhuis RML, Nederend M, Jongbloed MRM, et al. The potential of sodium-glucose cotransporter 2 inhibitors for the treatment of systemic right ventricular failure in adults with congenital heart disease. *Front Cardiovasc Med*. 2023;10:1093201. doi:10.3389/fcvm.2023.1093201

48. Fusco F, Scognamiglio G, Abbate M, et al. Dapagliflozin in patients with a failing systemic right ventricle: results from the DAPA-SERVE trial. *JACC Heart Fail*. 2024;12(4):789-791. doi:10.1016/j.jchf.2024.01.006

49. Neijenhuis RM, Regeer MV, Walker NL, et al. Effect of sodium-glucose cotransporter 2 inhibitors on ventricular function in systemic right ventricular failure. *Open Heart*. 2025;12(2):e003445. doi:10.1136/openhrt-2025-003445

50. Egorova AD, Nederend M, Tops LF, Vliegen HW, Jongbloed MRM, Kiès P. The first experience with sodium-glucose cotransporter 2 inhibitor for the treatment of systemic right ventricular failure. *ESC Heart Fail*. 2022;9(3):2007-2012. doi:10.1002/ehf2.13871

51. Gaydos SS, McHugh KE, Woodard FK, et al. Sodium-glucose cotransporter-2 inhibitor use in patients with a Fontan circulation. *Cardiol Young*. 2025;35(4):745-747. doi:10.1017/S1047951125000514

52. Konduri A, West C, Lowery R, et al. Experience with SGLT2 inhibitors in patients with single ventricle congenital heart disease and Fontan circulatory failure. *Pediatr Cardiol*. 2025;46(1):81-88. doi:10.1007/s00246-023-03332-5

53. Muneuchi J, Sugitani Y, Kobayashi M, Ezaki H, Yamada H, Watanabe M. Feasibility and safety of sodium glucose cotransporter-2 inhibitors in adults with heart failure after the Fontan procedure. *Case Rep Cardiol*. 2022;2022:5243594. doi:10.1155/2022/5243594

54. Neijenhuis RML, Regeer MV, Walker NL, et al. Echocardiographic effects of sodium-glucose cotransporter 2 inhibitors in single ventricle circulatory failure. *Int J Cardiol Congenit Heart Dis*. 2025;21:100603. doi:10.1016/j.ijcchd.2025.100603

55. Neijenhuis RML, MacDonald ST, Zemrak F, et al. Effect of sodium-glucose cotransporter 2 inhibitors in adults with congenital heart disease. *J Am Coll Cardiol*. 2024;83(15):1403-1414. doi:10.1016/j.jacc.2024.02.017

56. Bigelow AM, Riggs KW, Morales DLS, et al. Isosorbide dinitrate effect on hemodynamic profile, liver stiffness, and exercise tolerance in Fontan circulation (the NEET clinical trial). *Pediatr Cardiol*. 2024;45(7):1389-1397. doi:10.1007/s00246-023-03156-3

57. Waldmann V, Bessière F, Gardey K, et al. Systematic electrophysiological study prior to pulmonary valve replacement in tetralogy of Fallot: a prospective multicenter study. *Circ Arrhythm Electrophysiol*. 2023;16(6):e011745. doi:10.1161/CIRCEP.122.011745

58. Wallet J, Kimura Y, Blom NA, et al. Ventricular tachycardia substrates in children and young adults with repaired tetralogy of Fallot. *JACC Clin Electrophysiol*. 2024;10(12):2613-2624. doi:10.1016/j.jacep.2024.07.016

59. Noonan JA, Nadas AS, Rudolph AM, Harris GB. Transposition of the great arteries: a correlation of clinical, physiologic and autopsy data. *N Engl J Med*. 1960;263:592-596. doi:10.1056/NEJM196010062631406

60. Rashkind WJ, Miller WW. Creation of an atrial septal defect without thoracotomy: a palliative approach to complete transposition of the great arteries. *JAMA*. 1966;196(11):991-992. doi:10.1001/jama.1966.03100240125026

61. Co-Vu JG, Ginde S, Bartz PJ, Frommelt PC, Tweddell JS, Earing MG. Long-term outcomes of the neoaorta after arterial switch operation for transposition of the great arteries. *Ann Thorac Surg*. 2013;95(5):1654-1659. doi:10.1016/j.athoracsur.2012.10.081

62. Cedars A, Jacobs ML, Gottlieb-Sen D, et al. Reoperations in adolescents and adults after prior arterial switch operation: the Society of Thoracic Surgeons Congenital Heart Surgery Database analysis. *Ann Thorac Surg*. 2024;118(5):1080-1087. doi:10.1016/j.athoracsur.2024.05.038

63. Senning A. Surgical correction of transposition of the great vessels. *Surgery*. 1959;45(6):966-980.

64. Mustard WT. Successful two-stage correction of transposition of the great vessels. *Surgery*. 1964;55:469-472.

65. Ansari Ramandi MM, Budts W, Roos-Hesselink JW, et al. Heart failure in patients with a systemic right ventricle: a multicentre study with long-term follow-up. *Eur J Heart Fail*. 2025;27(12):3048-3057. doi:10.1002/ejhf.3664

66. Broberg CS, van Dissel A, Minnier J, et al. Long-term outcomes after atrial switch operation for transposition of the great arteries. *J Am Coll Cardiol*. 2022;80(10):951-963. doi:10.1016/j.jacc.2022.06.020

67. Jensen AS, Jørgensen TH, Christersson C, et al. Cause-specific mortality in patients during long-term follow-up after atrial switch for transposition of the great arteries. *J Am Heart Assoc*. 2022;11(14):e023921. doi:10.1161/JAHA.121.023921

68. Chaix MA, Dore A, Mondésert B, et al. Angiotensin receptor–neprilysin inhibitor vs placebo in congenital systemic right ventricular heart failure: the PARACYS-RV trial. *Eur Heart J*. 2024;45(16):1481-1483. doi:10.1093/eurheartj/ehad890

69. Angeli E, D'Angelo EC, Ragni L, et al. Candidacy for heart transplantation in adult congenital heart disease patients: a cohort study. *Int J Cardiol Congenit Heart Dis*. 2022;8:100363. doi:10.1016/j.ijcchd.2022.100363

70. Gallotti RG, Madnawat H, Shannon KM, Aboulhosn JA, Nik-Ahd F, Moore JP. Mechanisms and predictors of recurrent tachycardia after catheter ablation for d-transposition of the great arteries after the Mustard or Senning operation. *Heart Rhythm*. 2017;14(3):350-356. doi:10.1016/j.hrthm.2016.11.031

71. Amir R, Ladouceur M, Danford D, et al; AARCC and MARES Research Groups. Tricuspid valve surgery in transposition of the great arteries with a systemic right ventricle. *Int J Cardiol*. 2026;445:134055. doi:10.1016/j.ijcard.2025.134055

72. Jolley MA, Sulentic A, Amin S, et al. Introduction of transcatheter edge-to-edge repair in patients with congenital heart disease at a children's hospital. *Catheter Cardiovasc Interv*. 2024;103(2):326-334. doi:10.1002/ccd.30935

73. Vanpraagh R, Ongley PA, Swan HJ. Anatomic types of single or common ventricle in man: morphologic and geometric aspects of 60 necropsied cases. *Am J Cardiol*. 1964;13:367-386. doi:10.1016/0002-9149(64)90453-9

74. Khairy P, Poirier N, Mercier LA. Univentricular heart. *Circulation*. 2007;115(6):800-812. doi:10.1161/CIRCULATIONAHA.105.592378

75. Fontan F, Baudet E. Surgical repair of tricuspid atresia. *Thorax*. 1971;26(3):240-248. doi:10.1136/thx.26.3.240

76. D'Angelo EC, Ciuca C, Egidy Assenza G. Management of Fontan failure. *Heart*. 2022;108(22):1822-1831. doi:10.1136/heartjnl-2022-321006

77. Veldtman GR, Abualsaud A, Cohen S, et al. Fontan circulation and systemic disease—a retrospective cohort analysis over 35 years of follow-up. *Am Heart J*. 2025;279:40-49. doi:10.1016/j.ahj.2024.10.008

78. d'Udekem Y, Iyengar AJ, Galati JC, et al. Redefining expectations of long-term survival after the Fontan procedure: twenty-five years of follow-up from the entire population of Australia and New Zealand. *Circulation*. 2014;130(11)(suppl 1):S32-S38. doi:10.1161/CIRCULATIONAHA.113.007764

79. Gaebert P, Schaeffer T, Palm J, et al. Incidence, pathophysiology, and treatment of failing Fontan after the total cavopulmonary connection. *Cardiol Young*. 2024;34(11):2406-2413. doi:10.1017/S1047951124025782

80. Alsaied T, Rathod RH, Aboulhosn JA, et al. Reaching consensus for unified medical language in Fontan care. *ESC Heart Fail*. 2021;8(5):3894-3905. doi:10.1002/ehf2.13294

81. Rychik J, Atz AM, Celermajer DS, et al; American Heart Association Council on Cardiovascular Disease in the Young and Council on Cardiovascular and Stroke Nursing. Evaluation and management of the child and adult with Fontan circulation: a scientific statement from the American Heart Association. *Circulation*. 2019;140(6):e234-e284. doi:10.1161/CIR.0000000000000696

82. Emamaullee J, Zaidi AN, Schiano T, et al. Fontan-associated liver disease: screening, management, and transplant considerations. *Circulation*. 2020;142(6):591-604. doi:10.1161/CIRCULATIONAHA.120.045597

83. Goldberg DJ, Surrey LF, Glatz AC, et al. Hepatic fibrosis is universal following Fontan operation, and severity is associated with time from surgery: a liver biopsy and hemodynamic study. *J Am Heart Assoc*. 2017;6(5):e004809. doi:10.1161/JAHA.116.004809

84. Kim YY, Lluri G, Haeffele C, et al. Hepatocellular carcinoma in survivors after Fontan operation: a case-control study. *Eur Heart J*. 2024;45(16):1477-1480. doi:10.1093/eurheartj/ehad788

85. Khairy P, Fernandes SM, Mayer JE Jr, et al. Long-term survival, modes of death, and predictors of mortality in patients with Fontan surgery. *Circulation*. 2008;117(1):85-92. doi:10.1161/CIRCULATIONAHA.107.738559

86. Butts R, Atz AM, Baez Hernandez N, Sutcliffe D, Reisch J, Mahony L. Carvedilol does not improve exercise performance in Fontan patients: results of a crossover trial. *Pediatr Cardiol*. 2021;42(4):934-941. doi:10.1007/s00246-021-02565-6

87. Venkatesh P, Gao H, Abudayyeh I, Pai RG, Varadarajan P. Contemporary management of the failing Fontan. *J Clin Med*. 2024;13(11):3049. doi:10.3390/jcm13113049

88. Goldberg DJ, Zak V, Goldstein BH, et al; Pediatric Heart Network Investigators. Results of the FUEL trial. *Circulation*. 2020;141(8):641-651. doi:10.1161/CIRCULATIONAHA.119.044352

89. Deshaies C, Hamilton RM, Shohoudi A, et al; Alliance for Adult Research in Congenital Cardiology (AARCC). Thromboembolic risk after atriopulmonary, lateral tunnel, and extracardiac conduit Fontan surgery. *J Am Coll Cardiol*. 2019;74(8):1071-1081. doi:10.1016/j.jacc.2019.06.051

90. Freddo AM, Mondal A, Tomlinson AZ, et al. Comparison of arrhythmia prevalence and incidence in adult patients with lateral tunnel and extra-cardiac Fontan circulation. *Pediatr Cardiol*. 2026;47(4):1658-1665. doi:10.1007/s00246-025-03950-1

91. Quinton E, Nightingale P, Hudsmith L, et al. Prevalence of atrial tachyarrhythmia in adults after Fontan operation. *Heart*. 2015;101(20):1672-1677. doi:10.1136/heartjnl-2015-307514

92. Klehs S, Schneider HE, Backhoff D, Paul T, Krause U. Radiofrequency catheter ablation of atrial tachycardias in congenital heart disease: results with special reference to complexity of underlying anatomy. *Circ Arrhythm Electrophysiol*. 2017;10(12):e005451. doi:10.1161/CIRCEP.117.005451

93. Pundi KN, Pundi KN, Johnson JN, et al. Sudden cardiac death and late arrhythmias after the Fontan operation. *Congenit Heart Dis*. 2017;12(1):17-23. doi:10.1111/chd.12401

94. Mandalenakis Z, Rosengren A, Lappas G, et al. Atrial fibrillation burden in young patients with congenital heart disease. *Circulation*. 2018;137(9):928-937. doi:10.1161/CIRCULATIONAHA.117.029590

95. Brouwer C, Hebe J, Nürnberg JH, et al. Incidence of atrial fibrillation after radiofrequency catheter ablation for atrial tachycardia in congenital heart disease. *JACC Clin Electrophysiol*. 2024;10(6):1050-1060. doi:10.1016/j.jacep.2024.02.020

96. Moss AJ, Hall WJ, Cannom DS, et al; MADIT-CRT Trial Investigators. Cardiac-resynchronization therapy for the prevention of heart-failure events. *N Engl J Med*. 2009;361(14):1329-1338. doi:10.1056/NEJMoa0906431

97. Chubb H, Rosenthal DN, Almond CS, et al. Impact of cardiac resynchronization therapy on heart transplant–free survival in pediatric and congenital heart disease patients. *Circ Arrhythm Electrophysiol*. 2020;13(4):e007925. doi:10.1161/CIRCEP.119.007925

98. Zwaenepoel BAC, Neijenhuis RML, Veldtman G, et al. Effectiveness of cardiac resynchronization therapy in adult congenital heart disease: a meta-analysis of biventricular and conduction system pacing outcomes. *Heart Rhythm*. 2026;23(4):e624-e635. doi:10.1016/j.hrthm.2026.01.004

99. Fusco F, Scognamiglio G, Dellborg M, et al. Cardiac resynchronisation therapy among adults with a systemic right ventricle: a multicentre experience. *Heart*. 2026;112(10):549-556. doi:10.1136/heartjnl-2025-326384

100. Joyce J, O'Leary ET, Mah DY, Harrild DM, Rhodes J. Cardiac resynchronization therapy improves the ventricular function of patients with Fontan physiology. *Am Heart J*. 2020;230:82-92. doi:10.1016/j.ahj.2020.09.018

101. Jastrzębski M, Kiełbasa G, Cano O, et al. Left bundle branch area pacing outcomes: the multicentre European MELOS study. *Eur Heart J*. 2022;43(40):4161-4173. doi:10.1093/eurheartj/ehac445

102. Givertz MM, DeFilippis EM, Landzberg MJ, Pinney SP, Woods RK, Valente AM. Advanced heart failure therapies for adults with congenital heart disease: JACC state-of-the-art review. *J Am Coll Cardiol*. 2019;74(18):2295-2312. doi:10.1016/j.jacc.2019.09.004

103. Stehlik J, Edwards LB, Kucheryavaya AY, et al. The registry of the International Society for Heart and Lung Transplantation: twenty-seventh official adult heart transplant report–2010. *J Heart Lung Transplant*. 2010;29(10):1089-1103. doi:10.1016/j.healun.2010.08.007

104. Khush KK, Cherikh WS, Chambers DC, et al. The International Thoracic Organ Transplant Registry of the International Society for Heart and Lung Transplantation: thirty-fifth adult heart transplantation report–2018: focus theme: multiorgan transplantation. *J Heart Lung Transplant*. 2018;37(10):1155-1168. doi:10.1016/j.healun.2018.07.022

105. Christie JD, Edwards LB, Kucheryavaya AY, et al. The registry of the International Society for Heart and Lung Transplantation: twenty-seventh official adult lung and heart-lung transplant report–2010. *J Heart Lung Transplant*. 2010;29(10):1104-1118. doi:10.1016/j.healun.2010.08.004

106. Maxwell BG, Wong JK, Sheikh AY, Lee PH, Lobato RL. Heart transplantation with or without prior mechanical circulatory support in adults with congenital heart disease. *Eur J Cardiothorac Surg*. 2014;45(5):842-846. doi:10.1093/ejcts/ezt498

107. Cedars A, Vanderpluym C, Koehl D, Cantor R, Kutty S, Kirklin JK. An Interagency Registry for Mechanically Assisted Circulatory Support (INTERMACS) analysis of hospitalization, functional status, and mortality after mechanical circulatory support in adults with congenital heart disease. *J Heart Lung Transplant*. 2018;37(5):619-630. doi:10.1016/j.healun.2017.11.010

108. Marino BS, Lipkin PH, Newburger JW, et al; American Heart Association Congenital Heart Defects Committee, Council on Cardiovascular Disease in the Young, Council on Cardiovascular Nursing, and Stroke Council. Neurodevelopmental outcomes in children with congenital heart disease: evaluation and management: a scientific statement from the American Heart Association. *Circulation*. 2012;126(9):1143-1172. doi:10.1161/CIR.0b013e318265ee8a

109. Lee FT, Sun L, van Amerom JFP, et al. Fetal hemodynamics, early survival, and neurodevelopment in patients with cyanotic congenital heart disease. *J Am Coll Cardiol*. 2024;83(13):1225-1239. doi:10.1016/j.jacc.2024.02.005

110. Khanna AD, Duca LM, Kay JD, Shore J, Kelly SL, Crume T. Prevalence of mental illness in adolescents and adults with congenital heart disease from the Colorado Congenital Heart Defect Surveillance System. *Am J Cardiol*. 2019;124(4):618-626. doi:10.1016/j.amjcard.2019.05.023

111. Kovacs AH, Luyckx K, Thomet C, et al; APPROACH-IS Consortium, International Society for Adult Congenital Heart Disease. Anxiety and depression in adults with congenital heart disease. *J Am Coll Cardiol*. 2024;83(3):430-441. doi:10.1016/j.jacc.2023.10.043

112. De Backer J, Haugaa KH, Hasselberg NE, et al; ESC Scientific Document Group. 2025 ESC guidelines for the management of cardiovascular disease and pregnancy. *Eur Heart J*. 2025;46(43):4462-4568. doi:10.1093/eurheartj/ehaf193

113. Gurvitz M, Ionescu-Ittu R, Guo L, et al. Prevalence of cancer in adults with congenital heart disease compared with the general population. *Am J Cardiol*. 2016;118(11):1742-1750. doi:10.1016/j.amjcard.2016.08.057